

# Programming 4H-Silicon Carbide's Hamiltonian

Deval Deliwala

June 17, 2025

## Contents

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| <b>1</b> | <b>Zeeman Hamiltonian</b>                     | <b>2</b>  |
| <b>2</b> | <b>Hyperfine Hamiltonian</b>                  | <b>3</b>  |
| 2.1      | Two-electron + One-nucleus . . . . .          | 3         |
| 2.2      | Two-electron + Two-nuclei . . . . .           | 4         |
| 2.3      | Programming the Hamiltonian . . . . .         | 5         |
| 2.4      | Converting to Isotropic Hamiltonian . . . . . | 7         |
| <b>3</b> | <b>Zero-Field Splitting</b>                   | <b>9</b>  |
| <b>4</b> | <b>Exchange Hamiltonian</b>                   | <b>11</b> |
| <b>5</b> | <b>Basis Appendix</b>                         | <b>12</b> |
| 5.1      | Zeeman Basis . . . . .                        | 12        |
| 5.2      | Coupled Basis . . . . .                       | 13        |

In this paper I define the basis of  $\uparrow, \downarrow$ 's as the *Zeeman Basis*. I define the basis with the electron's coupled and the nucleus's still in the Zeeman basis as the *coupled basis*. I.e.

$$|\uparrow\uparrow\rangle|\uparrow\uparrow\rangle \iff |11\rangle|\uparrow\uparrow\rangle$$

defines a conversion between the *Zeeman* and *Coupled* bases.

See section 5 for a complete description of both bases.

# 1 Zeeman Hamiltonian

The Zeeman Hamiltonian is defined in the coupled-basis via

$$\hat{H}_Z|s, m\rangle|m_{I_1}\rangle|m_{I_2}\rangle = (mg_e\mu_B B_0 + m_{I_1}g_{n_1}\mu_N B_0 + m_{I_2}g_{n_2}\mu_N B_0)|s, m\rangle|m_{I_1}\rangle|m_{I_2}\rangle.$$

Hence, the coupled-basis is itself the eigenbasis. To build the Hamiltonian, we first define the coupled basis in the following manner:

```
1 import sympy as sp
2
3 """ build coupled-basis """
4
5 # electron |s, m> pairs
6 electron_states = [
7     (1, +1),
8     (1, 0),
9     (0, 0),
10    (1, -1),
11 ]
12
13 # nuclear |m_{I_1}>|m_{I_2}> pairs, s=1/2 implicit
14 nuclear_pairs = [
15     ( sp.Rational(+1,2), sp.Rational(+1,2) ), # (+1/2, +1/2)
16     ( sp.Rational(+1,2), sp.Rational(-1,2) ), # (+1/2, -1/2)
17     ( sp.Rational(-1,2), sp.Rational(+1,2) ), # (-1/2, +1/2)
18     ( sp.Rational(-1,2), sp.Rational(-1,2) ), # (-1/2, -1/2)
19 ]
```

Finally, we apply  $\hat{H}_Z$  to every basis state and generate the 16×16 Zeeman Hamiltonian via `sp.diag` since the coupled-basis is already the eigen (diagonal)-basis.

```
1 # zeeman frequencies
2 omega_e = g_e * mu_B * B0
3 omega_n1 = g_n1 * mu_N * B0
4 omega_n2 = g_n2 * mu_N * B0
5
6 # build the diagonal entries
7 entries = []
8 for s, m in electron_states:
9     for mI1, mI2 in nuclear_pairs:
10         entries.append(
11             m*omega_e + # electron zeeman
12             mI1*omega_n1 + # nucleus 1 zeeman
13             mI2*omega_n2 # nucleus 2 zeeman
14         )
15 H_Z = sp.diag(*entries) # Hamiltonian
```

This generates the final SymPy 16×16 Zeeman Hamiltonian.

## 2 Hyperfine Hamiltonian

### 2.1 Two-electron + One-nucleus

To understand how to define the Hyperfine Hamiltonian for a two-electron + two-nuclear system, we start with a simpler two-electron (a, b) + one-nucleus (I) system. In this system, the Hyperfine Hamiltonian is defined as

$$\begin{aligned}\hat{H}_{HF} &= \hat{S}_a \cdot A_a \cdot \hat{I} + \hat{S}_b \cdot A_b \cdot \hat{I} \\ &= \sum_{k \in \{a, b\}} A_{kx} S_{kx} I_x + A_{ky} S_{ky} I_y + A_{kz} S_{kz} I_z.\end{aligned}$$

Using ladder operator formalism, where

$$\hat{S}_x = \frac{1}{2} (\hat{S}_+ + \hat{S}_-) \quad \hat{S}_y = \frac{1}{2i} (\hat{S}_+ - \hat{S}_-), \quad (1)$$

and likewise for  $\hat{I}_x, \hat{I}_y$  acting on the nuclear states, we have

$$\begin{aligned}\hat{H}_{HF} &= \sum_k A_{kx} \left( \frac{1}{2} (S_{k+} + S_{k-}) \right) \left( \frac{1}{2} (\hat{I}_+ + \hat{I}_-) \right) \\ &\quad + A_{ky} \left( \frac{1}{2i} (S_{k+} - S_{k-}) \right) \left( \frac{1}{2i} (I_+ - I_-) \right) + A_{kz} S_{kz} I_z \\ &= \sum_k \frac{A_{kx}}{4} (S_{k+} I_+ + S_{k+} I_- + S_{k-} I_+ + S_{k-} I_-) \\ &\quad - \frac{A_{ky}}{4} (S_{k+} I_+ - S_{k+} I_- - S_{k-} I_+ + S_{k-} I_-) + A_{kz} S_{kz} I_z.\end{aligned}$$

And now since

$$\hat{S}_\pm |s, m\rangle = \hbar \sqrt{s(s+1) - m(m \pm 1)} |s, m \pm 1\rangle,$$

we can derive a full expression for  $\hat{H}_{HF} |m_a, m_b, m_I\rangle$ . Note  $s = 1/2$  is implicit and we are working in the Zeeman  $|\uparrow, \downarrow\rangle$  basis to make things easier for now.

We calculate an expression for  $H_{HF}$  that works for all 8 basis states in the two-electron + one-nucleus system. On the individual electron states, we see

$$\begin{aligned}S_+ |\uparrow\rangle &= 0 & S_+ |\downarrow\rangle &= \hbar |\uparrow\rangle \\ S_- |\uparrow\rangle &= \hbar |\downarrow\rangle & S_- |\downarrow\rangle &= 0 \\ S_z |m\rangle &= m\hbar |m\rangle, & m &\in \{\uparrow, \downarrow\}.\end{aligned}$$

and likewise for the nuclear state,

$$\begin{aligned}
I_+|\uparrow\rangle &= 0 & I_+|\downarrow\rangle &= \hbar|\uparrow\rangle \\
I_-|\downarrow\rangle &= 0 & I_-|\uparrow\rangle &= \hbar|\downarrow\rangle \\
I_z|m_I\rangle &= m_I\hbar|m_I\rangle, \quad m_I \in \{\uparrow, \downarrow\}.
\end{aligned}$$

Therefore, there are five nontrivial (that don't end up = 0) operator products:

| product     | nonzero for                      | result                                  |
|-------------|----------------------------------|-----------------------------------------|
| $S_{k+}I_+$ | $ \downarrow, \downarrow\rangle$ | $\hbar^2 \uparrow, \uparrow\rangle$     |
| $S_{k+}I_-$ | $ \downarrow, \uparrow\rangle$   | $\hbar^2 \uparrow, \downarrow\rangle$   |
| $S_{k-}I_+$ | $ \uparrow, \downarrow\rangle$   | $\hbar^2 \downarrow, \uparrow\rangle$   |
| $S_{k-}I_-$ | $ \uparrow, \uparrow\rangle$     | $\hbar^2 \downarrow, \downarrow\rangle$ |
| $S_{kz}I_z$ | any $m_{a,b}m_I$                 | $\hbar^2 m_{a,b}m_I m_{a,b}m_I\rangle$  |

Hence, using this table, we can construct the Hamiltonian:

$$\begin{aligned}
\hat{H}_{HF}|m_a, m_b, m_I\rangle &= \sum_{k \in \{a,b\}} \frac{\hbar^2}{4} (A_{kx} - A_{ky}) \hat{S}_{k+} \hat{I}_+ \delta_{m_k, m_I} | -m_k, -, -m_I\rangle \\
&\quad + \frac{\hbar^2}{4} (A_{kx} - A_{ky}) \hat{S}_{k+} \hat{I}_- \delta_{m_k, -m_I} | -m_k, -, -m_I\rangle \\
&\quad + A_{kz} \hbar^2 m_k m_I |m_k, -, m_I\rangle.
\end{aligned}$$

Expanding the summation,

$$\begin{aligned}
\hat{H}_{HF}|m_a, m_b, m_I\rangle &= \hbar^2 (A_{az} m_a m_I + A_{bz} m_b m_I) |m_a, m_b, m_I\rangle \\
&\quad + \frac{\hbar^2}{4} [(A_{ax} - A_{ay}) \delta_{m_a, m_I} + (A_{ax} + A_{ay}) \delta_{m_a, -m_I}] | -m_a, m_b, -m_I\rangle \\
&\quad + \frac{\hbar^2}{4} [(A_{bx} - A_{by}) \delta_{m_b, m_I} + (A_{bx} + A_{by}) \delta_{m_b, -m_I}] |m_a, -m_b, m_I\rangle.
\end{aligned}$$

where  $m_a, m_b, m_I$  all run over  $\{-\frac{1}{2}, \frac{1}{2}\}$ . The equation above defines the Hyperfine Hamiltonian in the Zeeman Basis for the full 8-basis-state two-electron + one-nuclear system. Evolving it to the two-electron + two-nuclear system is straightforward.

## 2.2 Two-electron + Two-nuclei

For two electrons  $k \in \{a, b\}$  and two nuclei  $p \in \{1, 2\}$ ,

$$\hat{H}_{HF} = \sum_{k \in \{a,b\}} \sum_{p=1}^2 (A_{kpx} S_{kx} I_{px} + A_{kpy} S_{ky} I_{py} + A_{kpz} S_{kz} I_{pz}).$$

Following the same procedure as the simpler system, we get the full Hyperfine Hamiltonian definition:

$$\begin{aligned}
& \hat{H}_{HF} |m_a, m_b, m_{I_1}, m_{I_2}\rangle \\
&= \hbar^2 \sum_{p=1}^2 (A_{apz} m_a m_{I_p} + A_{bpz} m_b m_{I_p}) |m_a, m_b, m_{I_1}, m_{I_2}\rangle \\
&+ \frac{\hbar^2}{4} [(A_{a1x} - A_{a1y}) \delta_{m_a, m_{I_1}} + (A_{a1x} + A_{a1y}) \delta_{m_a, -m_{I_1}}] | -m_a, m_b, -m_{I_1}, m_{I_2}\rangle \\
&+ \frac{\hbar^2}{4} [(A_{a2x} - A_{a2y}) \delta_{m_a, m_{I_2}} + (A_{a2x} + A_{a2y}) \delta_{m_a, -m_{I_2}}] | -m_a, m_b, m_{I_1}, -m_{I_2}\rangle \\
&+ \frac{\hbar^2}{4} [(A_{b1x} - A_{b1y}) \delta_{m_b, m_{I_1}} + (A_{b1x} + A_{b1y}) \delta_{m_b, -m_{I_1}}] |m_a, -m_b, -m_{I_1}, m_{I_2}\rangle \\
&+ \frac{\hbar^2}{4} [(A_{b2x} - A_{b2y}) \delta_{m_b, m_{I_2}} + (A_{b2x} + A_{b2y}) \delta_{m_b, -m_{I_2}}] |m_a, -m_b, m_{I_1}, -m_{I_2}\rangle.
\end{aligned} \tag{2}$$

However, remember this is in the Zeeman  $\uparrow, \downarrow$  basis. We need to convert it to the coupled triplet-singlet basis we used in the Zeeman Hamiltonian. Therefore, we apply a Clebsch-Gordan (CG) Unitary Transformation.

First we have

$$\begin{pmatrix} |\uparrow\uparrow\rangle \\ |\uparrow\downarrow\rangle \\ |\downarrow\uparrow\rangle \\ |\downarrow\downarrow\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} |11\rangle \\ |10\rangle \\ |00\rangle \\ |1-1\rangle \end{pmatrix},$$

defining a transformation between the electron Zeeman  $\rightarrow$  coupled basis. Therefore, defining

$$W = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where we  $\otimes I_4$  by the  $4 \times 4$  Identity because we leave the nuclear  $|m_{I_1}, m_{I_2}\rangle$  in the Zeeman basis and only couple the electrons. Then, once we have the Hyperfine Hamiltonian in the *Zeeman Basis*, we can apply the transformation

$$\hat{H}_{HF_{\text{coupled}}} = W \left( \hat{H}_{HF_{\text{Zeeman}}} \right) W^\dagger, \tag{3}$$

to get the final Hyperfine Hamiltonian in the correct coupled basis.

## 2.3 Programming the Hamiltonian

We first store Equation 2 as a `SymPy`  $16 \times 16$  matrix, which is in the Zeeman  $|m_a, m_b, m_{I_1}, m_{I_2}\rangle$  Basis.

We also have to ensure that after the CG transformation, shown in Equation 3, the now matrix coupled-basis ordering is the identical to the coupled-basis ordering of the Zeeman Hamiltonian. Maintaining the order of the coupled-basis is crucial.

The following code generates the 16 Zeeman basis states in correct order:

```

1 def _generate_zeeman_basis(self):
2     # Builds the Zeeman |m_a, m_b, m_I1, m_I2> basis with m = +/-1/2.
3
4     half = sp.Rational(1, 2)
5     return [
6         (m_a, m_b, m_I1, m_I2)
7         for m_a in (half, -half)
8         for m_b in (half, -half)
9         for m_I1 in (half, -half)
10        for m_I2 in (half, -half)
11    ]
12 ...
13 self.basis_ze = self._generate_zeeman_basis()

```

Now, to emulate Equation 2, we write the following functions:

```

1 def _delta_parallel(self, m_e, m_I):
2     return m_e == m_I
3
4 def _delta_antiparallel(self, m_e, m_I):
5     return m_e == -m_I
6
7 def _action_on_ket(self, ket):
8     # Calculates H_HF |ket> in Zeeman basis, returns dict {new_ket: coeff}.
9
10    m_a, m_b, m_I1, m_I2 = ket
11    psi = {}
12
13    # diagonal S_z I_z term: A * m_e * m_I
14    diag = (
15        self.A['AA1Z'] * m_a * m_I1 +
16        self.A['AB1Z'] * m_b * m_I1 +
17        self.A['AA2Z'] * m_a * m_I2 +
18        self.A['AB2Z'] * m_b * m_I2
19    )
20    psi[ket] = diag
21
22    # off-diagonal flip-flop S_x I_x + S_y I_y -> (A_x - A_y)/4 etc.
23    def add_flip(m_e, m_I, x_key, y_key, new_ket):
24        if self._delta_parallel(m_e, m_I):
25            coeff = (self.A[x_key] - self.A[y_key]) / 4
26            psi[new_ket] = coeff
27        elif self._delta_antiparallel(m_e, m_I):
28            coeff = (self.A[x_key] + self.A[y_key]) / 4
29            psi[new_ket] = coeff
30
31    add_flip(m_a, m_I1, 'AA1X', 'AA1Y', (-m_a, m_b, -m_I1, m_I2))
32    add_flip(m_a, m_I2, 'AA2X', 'AA2Y', (-m_a, m_b, m_I1, -m_I2))
33    add_flip(m_b, m_I1, 'AB1X', 'AB1Y', (m_a, -m_b, -m_I1, m_I2))
34    add_flip(m_b, m_I2, 'AB2X', 'AB2Y', (m_a, -m_b, m_I1, -m_I2))
35

```

```
36     return psi
```

`_delta_parallel` and `_delta_antiparallel` are the delta functions shown in Equation 2 and allow us which of  $\frac{1}{4}(A_{kpx} \pm A_{kpy})$  are the coefficients of the superposed kets.

We now perform the Hyperfine Hamiltonian (still Zeeman basis) on every Zeeman basis-state, building the full Hyperfine Zeeman Matrix as we go.

```
1 def _build_zeeman_matrix(self):
2     # Return the full 16x16 hyperfine template in the Zeeman basis.
3
4     size = len(self.basis_ze)
5     H = sp.MutableDenseMatrix(size, size, lambda _: 0)
6     for j, ket in enumerate(self.basis_ze):
7         action = self._action_on_ket(ket)
8         for new_ket, coeff in action.items():
9             i = self.index_ze[new_ket]
10            H[i, j] = coeff
11     return H.as_immutable()
12 ...
13 self.H_ze = self._build_zeeman_matrix()
```

Now all that is left is to apply the CG transformation (Equation 3). We first build the  $W$  via defining the  $4 \times 4$  CG matrix and taking the tensor product with the  $4 \times 4$  Identity:

```
1 def _build_cg_unitary(self):
2     # Builds the unitary for two-electron Clebsch-Gordan transform otimes identity
3     on nuclei.
4
5     half = sp.sqrt(sp.Rational(1, 2))
6     U = sp.Matrix([
7         [1, 0, 0, 0],
8         [0, half, half, 0],
9         [0, half, -half, 0],
10        [0, 0, 0, 1]
11    ])
12    I4 = sp.eye(4)
13    return sp.kronecker_product(U, I4)
14 ...
15 self.W = self._build_cg_unitary()
```

Finally, we get the Hyperfine Hamiltonian in the same coupled-basis as the Zeeman Hamiltonian:

```
1 self.H_coup = sp.simplify(self.W * self.H_ze * self.W.H) # Hyperfine Hamiltonian
```

## 2.4 Converting to Isotropic Hamiltonian

We have the full, correct, Hyperfine Hamiltonian. However, for computational feasibility, we will assume the hyperfine effect is *isotropic*, that is, we will assume

$$\begin{aligned} A_{a1x} &= A_{a1y} = A_{a1} \\ A_{b1x} &= A_{b1y} = A_{b1}, \end{aligned}$$

and likewise for the second nuclear hyperfine interaction. Hence, we implement

```

1 def convert_to_isotropic(H_sym: sp.Matrix) -> sp.Matrix:
2     # Substitute anisotropic A_xyz -> isotropic Aa, Ab values.
3
4     subs_map = {
5         A_a1x: Aa1, A_a1y: Aa1, A_a1z: Aa1,
6         A_a2x: Aa2, A_a2y: Aa2, A_a2z: Aa2,
7         A_b1x: Ab1, A_b1y: Ab1, A_b1z: Ab1,
8         A_b2x: Ab2, A_b2y: Ab2, A_b2z: Ab2,
9     }
10    return sp.simplify(H_sym.subs(subs_map))

```

This returns the final isotropic Hyperfine Hamiltonian we will be using for analysis.



### 3 Zero-Field Splitting

The Zero-Field Splitting (ZFS) Hamiltonian, also known as the Dipole-Dipole Hamiltonian, is defined by

$$\hat{H}_{ZFS} = D_1 \left( \hat{S}_z^2 - \frac{1}{3} (\hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2) \right) + D_2 (\hat{S}_x^2 - \hat{S}_y^2). \quad (4)$$

Using raising/lowering operator formalism again, we have

$$\begin{aligned} \hat{S}_+ |s, m\rangle &= \frac{1}{2} (\hat{S}_x + i\hat{S}_y) |s, m\rangle = \sqrt{s(s+1) - m(m+1)} |s, m+1\rangle \\ \hat{S}_- |s, m\rangle &= \frac{1}{2} (\hat{S}_x - i\hat{S}_y) |s, m\rangle = \sqrt{s(s+1) - m(m-1)} |s, m-1\rangle. \end{aligned}$$

Hence, we retrieve Equation 1 and find

$$\begin{aligned} \hat{S}_x &= \frac{1}{2} (\hat{S}_+ + \hat{S}_-) \\ \hat{S}_y &= \frac{1}{2i} (\hat{S}_+ - \hat{S}_-) \\ \hat{S}_x^2 + \hat{S}_y^2 &= \hat{S}_x^2 - \hat{S}_y^2 = \frac{1}{2} (\hat{S}_+^2 + \hat{S}_-^2). \end{aligned}$$

Substituting this into Equation 4, we get

$$\hat{H}_{ZFS} |s, m\rangle = D_1 m^2 |s, m\rangle - \frac{D_1}{3} (s(s+1)) |s, m\rangle \quad (5)$$

$$+ \frac{D_2}{2} (s(s+1) - m(m+1)) |s, m+2\rangle \quad (6)$$

$$+ \frac{D_2}{2} (s(s+1) - m(m-1)) |s, m-2\rangle, \quad (7)$$

where the last two terms highlight the forbidden  $m \pm 2$  transitions that give rise to the half-field response. This Hamiltonian acts on the coupled basis, akin to the Zeeman Hamiltonian. Hence, we translate the above ZFS Hamiltonian definition as follows:

```

1 electron_states = [(1, 1), (1, 0), (0, 0), (1, -1)]
2
3 # build 4x4 electron ZFS Hamiltonian
4 H_elec = sp.zeros(4)
5 for i, (s, m) in enumerate(electron_states):
6     # Diagonal term: D1 * m^2 - (D1/3) * s*(s+1)
7     H_elec[i, i] = D1*m**2 - (D1/3)*s*(s+1)
8     # Off-diagonal terms coupling m -> m+/-2
9     for dm, expr in [(2, D2/2*(s*(s+1) - m*(m+1))),
10                      (-2, D2/2*(s*(s+1) - m*(m-1)))]:
11         newm = m + dm
12         if (s, newm) in electron_states:
13             j = electron_states.index((s, newm))
14             H_elec[i, j] = expr

```

```

15
16 # 4x4 identity nuclear ZFS Hamiltonian
17 I_nuc = sp.eye(4)
18
19 # Final ZFS Hamiltonian
20 H_ZFS = sp.kronecker_product(H_elec, I_nuc)

```

where we again take the tensor product  $H_{\text{electron ZFS}} \otimes I_4$  to preserve basis ordering. This gives the final ZFS/Dipole-Dipole Hamiltonian in the correct coupled-basis.

## 4 Exchange Hamiltonian

The isotropic Exchange Hamiltonian between electrons is defined by

$$\hat{H}_{EX} = -J\hat{S}_a \cdot \hat{S}_b,$$

where we add a  $-1$  factor to yield  $-\frac{J}{4}$  for triplet states and  $+\frac{3J}{4}$  for singlet states. Therefore, its action on a coupled  $|s, m\rangle$  state of two electrons is given by

$$\hat{H}_{EX}|s, m\rangle = -J\left(\hat{S}_a \cdot \hat{S}_b\right)|s, m\rangle = -J\left(\frac{s(s+1) - \frac{3}{2}}{2}\right)|s, m\rangle.$$

Because this already acts on the coupled-basis (nuclear  $|m_I\rangle$  are left alone), programming this Hamiltonian is the easiest.

```

1 # same electron basis ordering
2 electron_states = [(1, 1), (1, 0), (0, 0), (1, -1)]
3
4 # build 4x4 electron exchange Hamiltonian
5 H_ex_elec = sp.zeros(4)
6 for i, (s, m) in enumerate(electron_states):
7     H_ex_elec[i, i] = -J * (s * (s + 1) - 1.5) / 2
8
9 # 4x4 nuclear identity
10 I_nuc = sp.eye(4)
11
12 # Exchange Hamiltonian
13 H_EX = sp.kronecker_product(H_ex_elec, I_nuc)

```

Here, to generate the full  $16 \times 16$  Exchange Hamiltonian that acts on the full two-electron + two-nuclei system, we again take the tensor product of the  $4 \times 4$  electron Exchange Hamiltonian with the  $4 \times 4$  nuclear Identity matrix to yield the full Exchange Hamiltonian on the system.

The Full 4H-Silicon Carbide Spin Hamiltonian is finally,

```

1 H_SPIN = H_zeeman + H_hyperfine + H_zfs + H_exchange.

```

I.e., the sum of all previously defined Hamiltonians defined in same coupled-basis throughout.

## 5 Basis Appendix

### 5.1 Zeeman Basis

The Zeeman-Basis used for calculating the Hyperfine Hamiltonian is defined by

|    |                                                                  |
|----|------------------------------------------------------------------|
| 0  | $ +\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 1  | $ +\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 2  | $ +\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 3  | $ +\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 4  | $ +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 5  | $ +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 6  | $ +\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 7  | $ +\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 8  | $ -\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 9  | $ -\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 10 | $ -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 11 | $ -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 12 | $ -\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 13 | $ -\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 14 | $ -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 15 | $ -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\rangle$ |

where  $|+\frac{1}{2}\rangle = |\uparrow\rangle$  and  $-\frac{1}{2} = |\downarrow\rangle$  (electron) or  $|\uparrow, \downarrow\rangle$  (nuclear)

Indices 0-15 represent matrix left-to-right or top-to-bottom basis states.

## 5.2 Coupled Basis

The Coupled-Basis used for calculating all individual Hamiltonians and summing them together to generate the final Spin Hamiltonian is defined by

|    |                 |           |                                      |
|----|-----------------|-----------|--------------------------------------|
| 0  | $ 1, 1\rangle$  | $\otimes$ | $ +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 1  | $ 1, 0\rangle$  | $\otimes$ | $ +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 2  | $ 0, 0\rangle$  | $\otimes$ | $ +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 3  | $ 1, -1\rangle$ | $\otimes$ | $ +\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 4  | $ 1, 1\rangle$  | $\otimes$ | $ +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 5  | $ 1, 0\rangle$  | $\otimes$ | $ +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 6  | $ 0, 0\rangle$  | $\otimes$ | $ +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 7  | $ 1, -1\rangle$ | $\otimes$ | $ +\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 8  | $ 1, 1\rangle$  | $\otimes$ | $ -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 9  | $ 1, 0\rangle$  | $\otimes$ | $ -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 10 | $ 0, 0\rangle$  | $\otimes$ | $ -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 11 | $ 1, -1\rangle$ | $\otimes$ | $ -\frac{1}{2}, +\frac{1}{2}\rangle$ |
| 12 | $ 1, 1\rangle$  | $\otimes$ | $ -\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 13 | $ 1, 0\rangle$  | $\otimes$ | $ -\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 14 | $ 0, 0\rangle$  | $\otimes$ | $ -\frac{1}{2}, -\frac{1}{2}\rangle$ |
| 15 | $ 1, -1\rangle$ | $\otimes$ | $ -\frac{1}{2}, -\frac{1}{2}\rangle$ |

where  $|+\frac{1}{2}\rangle = |\uparrow\rangle$  and  $-\frac{1}{2} = |\downarrow\rangle$  (nuclear-only).

Indices 0-15 represent matrix left-to-right or top-to-bottom basis states.